Raman Kumar Jha

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Education

Texas A&M University — College Station, USA
Doctor of Philosophy in Computer Engineering, — Aug 2026 – May 2030
Relevant Coursework: Multi Robot Systems, Deep Learning, Multimodal Computer Vision

New York University — New York, USA
Master of Science in Computer Engineering, GPA 3.7/4 — Sept 2024 – May 2026
Relevant Coursework: Foundation of Robotics, Robot Perception, Reinforcement Learning & Optimal Control for Robotics, Embodied Learning & Vision, Deep Learning; Plan, Learn & Control Autonomous Space Robots

Savitribai Phule Pune University — Pune, India
Bachelor of Engineering in Computer Engineering, GPA 8.8/10 — Jul 2019 – Jun 2023
Relevant Coursework: Principles of Programming Languages, Data Structures & Algorithms; Discrete Mathematics, Artificial Intelligence, Machine Learning, Data Science & Big Data Analytics, Deep Learning, Image Processing

Technical Skills

Programming: Python, C/C++, Matlab, Octave, LaTeX
ML/AI Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Keras, Stable Baselines3
Simulation & Tools: RViz, Gazebo, Isaac, Habitat, Linux, Git/GitHub, Docker, Anaconda, HuggingFace
Robotics & Perception: ROS, SLAM, PCL, Sensor Fusion, Camera Calibration, Motion Planning
LLM & NLP: LangChain, PEFT (LoRA/QLoRA), RAG, NLTK, Prompt Engineering, Multimodal Learning
Data Analysis: NumPy, Pandas, Matplotlib, SciPy, Seaborn
Cloud, and MLOps: AWS, GCP, Azure, Hugging Face, Wandb

Publications

[1] **VCS-SLAM: Visibility-Consistent and Surface-Coupled Semantic SLAM**
Raman Jha, Shuaihang Yuan, Yi Fang.
Robotics and Automation Letters(RAL) 2026 Paper

[2] **Adaptive Keyframe Selection for Scalable 3D Scene Reconstruction in Dynamic Environments.**
Raman Jha, Yang Zhou, Giuseppe Loianno.
International Conference on Robotics, Computer Vision and Intelligent Systems (ROBOVIS) 2026, Paper.

[3] **RT-X NEt: Cross Attention Network using RGB and Thermal Images for Low-Light Image Enhancement.**
Raman Jha, Adithya Lenka, Mani Ramanagopal, Aswin C. Sankaranarayanan, Kaushik Mitra.
International Conference on Image Processing (ICIP) 2025, Paper.

[4] **A Contrastive Meta-Learning Approach with Isotropic Sparse Decomposition for Scalable Audio-Visual Learning.**
Dhruv Dixit, Paritosh Pandey, **Raman Jha**, Pranshul Bhatnagar, Shashank Mouli Satapathy.
International Conference on Computing, Communication and Learning (CoCoLe) 2024, Paper.

[5] **Detection of Vehicle Lights and Classification of Its State.**
Raman Kumar Jha
International Conference on Advanced Computing Technologies and Applications (ICACTA) 2023, Paper.

[6] **Prediction of Bankruptcy of a Company Using Machine Learning Techniques.**
V. Kothuru, **Raman Kumar Jha**, S. Ranjit, B. V. Kumar Reddy, S. Roy, S. Sudheer.
International Conference on Electronics and Sustainable Communication Systems (ICESC) 2022, Paper.

Experience

Embodied AI and Robotics Lab, New York University — New York, USA
Research Assistant(MS Thesis) — Aug 2025 – May 2026

- Mentored by **Shuaihang Yuan**, and advised by **Prof. Yi Fang for MS thesis**.

- Addressed robot object goal navigation and semantic mapping challenges in visually complex environments.
- Developed a physically consistent semantic mapping module to mitigate semantic ghosts and occlusion leakage.
- Implemented a visibility-aware model and surface-coupled regularization to align semantic boundaries.
- Enforced temporal consistency and visibility gating to reduce the Occlusion Leakage Rate (OLR) and enhance spatial reasoning for robust autonomous navigation.

Agile Robotics and Perception Lab, New York University — New York, USA
Research Assistant — Oct 2024 – July 2025

- Mentored by **Yang Zhou** and supervised by **Prof. Giuseppe Loianno**.
- Worked on adaptive keyframe selection for scene reconstruction in dynamic environments.
- Developed a Transformer-based architecture to regress 3D point maps directly from 2D images.
- Optimized keyframe selection by implementing an error-based metric and momentum-based keyframe control.
- Reduced the required number of keyframes by 5% while improving the system's reconstruction on a drone.

Computational Imaging Lab, IIT Madras — Chennai, India
Research Associate — Aug 2023 – June 2024

- Advised by **Prof. Kaushik Mitra** and **Prof. Aswin C. Sankaranarayanan**.
- Developed a cross-attention network for low-light image enhancement and restoration in autonomous systems.
- Used thermal images as a guide for RGB images to improve the performance of the enhancement algorithm.
- Curated and released the V-TIEE dataset, consisting of 50 paired real-world RGB-Thermal sequences.
- Achieved 2 dB performance increase over traditional RGB-only networks for low-light image enhancement.

SeiSei.ai — Jaipur, India
Founding Machine Learning Intern — April 2023 – July 2023

- Designed a Generative AI pipeline for automated video transformation, visual face dubbing, and voice conversion.
- Created a system to detect and extract the landmarks from the face of a particular individual in a video.
- Implemented a facial landmark detection system to drive a GAN-based FreeVC model for voice cloning and a DINet model for visual dubbing.
- Improved inference performance by 17% and successfully integrated the pipeline into the company's production.

Dalhousie University — Halifax, Canada
Mitacs Globalink Research Intern | *Research Report* — July 2022 – Oct 2022

- Supervised by **Prof. Andrew Rutenberg**.
- Worked on classifying health variables into low and high dimensions for predicting aging and death.
- Used Variational Autoencoders and Recurrent Neural Networks for model development and testing.
- Trained and tested on the English Longitudinal Study of Ageing (ELSA) dataset.
- Developed latent space models improving performance by 14% and designed a modified separate filter to calculate the loss function for each health variable.

Centre of Visual Information Technology Lab (CVIT), IIIT Hyderabad — Hyderabad, India
Applied Research Fellow — Jan 2022 – June 2022

- Supervised by **Prof. Ravi Kiran Sarvadevabhatla** and **Prof. C V Jawahar**.
- Worked on detecting vehicle lights and classifying their states by training multiple object detection models, including YOLOv7, YOLOv5, YOLOv4, Faster R-CNN, and MobileNet.
- Used Indian Driving Dataset (IDD) for training and testing of the detection and classification system.
- YOLO V7 achieved best results with 88.4% mAP and was fine-tuned to improve detection of small lights.
- Applied color thresholding principle for the red and white light recognition in diverse environments.

Projects

Lunar Rover Navigation & Control | GitHub — Oct 2025 – Dec 2025

- Simulated a six-wheeled lunar rover for autonomous navigation in a GPS-denied, low-gravity environment.
- Targeted robust waypoint traversal while maintaining chassis stability and mitigating risks on uneven terrain.
- Implemented an EKF-based sensor fusion for state estimation and designed a stability-aware proportional control.
- Maintained a 0.6–0.9 stability margin and quantified a 55m localization drift caused by wheel slip in lunar regolith.

Self-Drive: Virtual Integrated Project | GitHub — Jan 2025 – May 2025

- Planned an autonomous driving system for real-world navigation in complex maze, and in real-world.
- Designed and integrated perception, planning, and control modules for robust decision-making.
- Utilized DINOv2-based lane detection and object recognition, and optimized A* for efficient path planning.
- Achieved 95% navigation accuracy in real-world maze environments with safe and reliable performance.

Diffusion-Based Model Predictive Control | GitHub — Feb 2025 – May 2025

- Designed a control strategy to improve robotic trajectory optimization in uncertain and dynamic environments.

- Enhanced MPC by integrating probabilistic modeling to handle uncertainty in unseen conditions.
- Combined diffusion models with MPC and implemented probabilistic trajectory generation using diffusion.
- Achieved a 20% improvement in trajectory optimization performance over traditional MPC approaches.

CLIP Model Benchmarking on Flickr8K | GitHub — Apr 2025

- Designed CLIP vision-language models for zero-shot image retrieval and classification on Flickr8K dataset.
- Objective was to assess how encoders impact retrieval quality and response to both concrete and abstract prompts.
- Constructed and pretrained three model pipelines, tested on custom prompts to evaluate performance.
- Larger models excelled at retrieval tasks, however smaller ones trained faster but ,performed worse on queries.

LoRA Fine-Tuning for Text Classification | GitHub — Mar 2025

- Adapted a pre-trained transformer (RoBERTa) for a text classification task in a resource-constrained environment.
- Aimed to maximize classification accuracy while minimizing additional parameter usage using PEFT techniques.
- Fine-tuned RoBERTa with LoRA adapters using low learning rate (1e-4), weight decay, and optimized batch size.
- Achieved 94.06% validation accuracy (final loss: 0.16) and 85.9% Kaggle test accuracy on final competition.

2D Quadrotor Control Using Reinforcement Learning | GitHub — Oct 2024 – Dec 2024

- Designed a control solution for a 2D quadrotor to reach target positions while avoiding obstacles.
- Implemented a reinforcement learning framework for stable navigation under dynamic constraints.
- Trained a PPO-based policy and engineered rewards balancing target accuracy, stability, and obstacle avoidance.
- Achieved robust flight control with smooth trajectories, minimal oscillations, and reliable obstacle avoidance.

American Sign Language Recognizer | GitHub — Jun 2022 – Jul 2022

- Built an American Sign Language recognizer using real-time hand gesture detection.
- Applied the YOLOv5 object detection framework to real-time camera images for accurate sign recognition.
- Recognized five distinct American Sign Language signs, including Hello, I Love You, Yes, No, and Thank You.
- Achieved 97.4% recognition accuracy across all American Sign Language sign classes during real-time camera-based testing and evaluation.

Driver Drowsiness Detector | GitHub — May 2021 – Jun 2021

- Developed a driver drowsiness detector based on real-time eye-state analysis.
- Trained a CNN-based model on real-time camera image sequences to detect signs of drowsiness.
- Classified the driver's state as awake or drowsy using extracted eye condition features and blink patterns.
- Achieved 85.2% detection accuracy for driver drowsiness classification across real-time camera-based testing and evaluation.

Teaching Experience

Teaching Assistant — Spring 2026
Computer Vision for Sciences and Engineering — New York University

- Assisted students with paper readings, research critiques, and developing effective research presentation skills.
- Mentored students through structured project guidance, including proposals and feasibility studies.
- Evaluated weighted presentations, peer-review assignments, and initial and final project reports.

Teaching Assistant — Fall 2025
Advanced Topics in Computer Vision — New York University

- Assisted students with their presentations, paper reviews, project queries, and coursework.
- Conducted office hours for the students to help them with their projects in Computer Vision.
- Graded student presentations, paper reviews, and course projects submitted through Brightspace.

Teaching Assistant — Spring 2025
Physics Informed Machine Learning — New York University

- Mentored students with their homework, quizzes, and projects, and addressed doubts on Piazza.
- Held office hours and tutorial sessions for the students in Physics-Informed Machine Learning.
- Evaluated student homework, quizzes, assignments, and midterms for the course on Brightspace.

Teaching Assistant — Spring 2025
Chemical Engineering Computation — New York University

- Assisted students with their homework, assignments, and in-class doubts on Python programming.
- Held office hours and guided students through their mini projects and final course projects.
- Graded homework, quizzes, assignments, and midterms for Chemical Engineering Computation on Brightspace.

Teaching Assistant — Spring 2024
Modern Computer Vision — IIT Madras

- Conducted tutorial classes on the basics of programming and deep learning for students.
- Evaluated student assignments and projects for the computer vision course through Moodle.
- Mentored students on their course projects, competitions, and hackathon participation.

Teaching Assistant Spring 2023
Machine Learning Savitribai Phule Pune University

- Graded student assignments, projects, and exams for the machine learning course.
- Held office hours and assisted students with their homework, projects, and hackathons.
- Assisted students and answered their questions on Sodl about reports and final projects.

Achievements and Extracurriculars

1 **Judge**, Hackathon LLUS X PULSE hosted by Korean International Students Organization 04/2026

2 **Winner**, Spark Hackathon hosted by NVIDIA, and sponsored by ACER 04/2026

3 **Outstanding Reviewer**, British Machine Vision Conference (BMVC) 11/2024

4 **Recipient**, merit-based scholarship of $7000 per year from New York University 04/2024

5 **Finalist**, Smart India Hackathon 08/2022

6 **Recipient**, Mitacs Globalink Research Fellowship 03/2022

7 **Winner**, Codeheat contest, FOSSASIA 09/2021

8 **Recipient**, Script Fellowship, Script organization 08/2021

Reviewer Experience

Reviewer: BMVC 2026, NeurIPS 2025, BMVC 2025, ICLR 2024, BMVC 2024, ICVGIP 2023

Leadership and Extra-curriculars

1 **Treasurer**, Hindu Student Union NYU (July 2025 – May 2026). Directed financial operations, including budget forecasting and allocation, while securing resources for diverse events to promote community engagement.

2 **Treasurer**, IEEE NYU Student Branch (Jun 2025 – May 2026). Managed financial activities, budget planning, and coordinated funding for technical events and workshops for the club.

2 **Steward**, GSOC NYU (May 2025 – May 2026). Represented as a steward for NYU Tandon School of Engineering, and helped students with queries, grievances, and issues related to their on-campus jobs.

3 **Mitacs Globalink Ambassador**, Mitacs Globalink Research Internship (Mar 2023 – Jun 2023). Assisted and guided upcoming research interns regarding the Mitacs Globalink Research Fellowship.

4 **Machine Learning Lead**, Google Developer Student Club (Aug 2021 – Jul 2022). Conducted events, seminars, tutorials, and guided juniors for research in Machine Learning.